

My Golem Electron App

This project packages a Golem-based R Shiny application inside an Electron desktop app for macOS, and it also builds the images for Linux and Windows.

It bundles the R runtime (**R.framework**) directly in the app, so no external R installation is required.

Project Structures

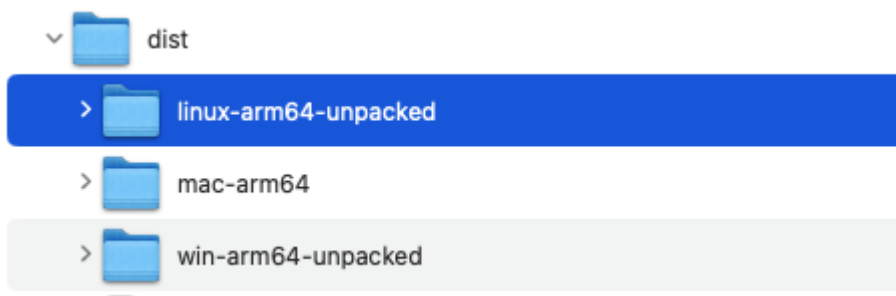
```
/electron
├── app/                                # Production R script for launching
Golem app
├──   └── run_app.R
├──   main.js                          # Electron main process
├──   resources/R.framework           # Bundled R runtime (flattened for
macOS)
├──   package.json                    # Electron build configuration
├──   scripts/                        # Build helper scripts
├──       ├── pre_build_fix_rframework.sh
├──       └── post_build_check.sh
```

Final Run

Build and deploy

- cd electron
- npm run prod-build

-
- cd dist
 - /dist folder contains the dmg to be installed.
 - install .dmg and run the app



- Launch the app

About This App

This is a modern Shiny app packaged inside Electron.

It uses shinythemes + shinyWidgets for a cleaner UI.

Electron + Shiny App

Dashboard

Settings

About

Choose dataset:

mtcars

Refresh Data

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
21.00	6.00	160.00	110.00	3.90	2.62	16.46	0.00	1.00	4.00	4.00
21.00	6.00	160.00	110.00	3.90	2.88	17.02	0.00	1.00	4.00	4.00
22.80	4.00	108.00	93.00	3.85	2.32	18.61	1.00	1.00	4.00	1.00
21.40	6.00	258.00	110.00	3.08	3.21	19.44	1.00	0.00	3.00	1.00
18.70	8.00	360.00	175.00	3.15	3.44	17.02	0.00	0.00	3.00	2.00
18.10	6.00	225.00	105.00	2.76	3.46	20.22	1.00	0.00	3.00	1.00

Build Process - Details

1. Prepare R.framework

This step flattens **R.framework** (removes **Versions/** symlinks, ensures **PrivateHeaders**, **Headers**, **Libraries**, **Resources** are real folders).

```
cd electron
npm run fix-rframework
```

Verify:

```
ls electron/resources/R.framework/Resources/bin/Rscript
```

It should exist.

2. Ensure **run_app.R** is bundled

Place your production launcher script in:

```
electron/app/run_app.R
```

It will be copied into:

```
Contents/Resources/app/run_app.R
```

inside the `.app` at build time.

3. Build the app

```
cd electron  
npm run prod-build
```

This runs:

- `fix-rframework`
- `electron-builder`
- `post-build-check` (verifies `Rscript` inside `.app`)

Output:

- `.dmg` and `.zip` in `electron/dist/`
-

Running the App

- Install `.dmg`
 - Open `My Golem Electron` from Applications
 - Electron will:
 1. Launch `Rscript run_app.R` from `Contents/Resources/app`
 2. Start Shiny server on fixed port (e.g. `4242`)
 3. Open `http://127.0.0.1:4242` in Electron window
-

Debugging

Check inside the `.app` bundle:

```
tree -L 3 dist/mac/YourApp.app/Contents/Resources
```

You should see:

```
Resources/
├── app/
│   └── run_app.R
├── R.framework/
│   └── Resources/
│       ├── bin/Rscript
│       └── main.js
```

1. Directly from the .app bundle

open `dist/mac-arm64/idepGolemPackage.app`, just double-click it in Finder.

2. From the .dmg

If you built a DMG, open it in Finder → drag the app to `/Applications` → then launch it from Launchpad.

3. From the command line inside the bundle (for debugging)

`dist/mac-arm64/idepGolemPackage.app/Contents/MacOS/idepGolemPackage`

This way you can see stdout/stderr in your terminal while R starts up.

Logs

`dist/mac-arm64/idepGolemPackage.app/Contents/Resources/golem-electron-debug-.log`